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COMPUTER SCIENCE PROGRAM

Bachelor's Thesis

INLINING COMPILER OPTIMIZATION THROUGH MACHINE LEARNING TECHNIQUES

Graduate

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Abstract

In optimizing compilers, the problem of inlining has long been studied in order to increase performance across platforms while keeping the size of applications small. Choosing when to inline is complicated because, in the context of a program, functions are interconnected through the call graph, and one decision can influence the structure of the program and future optimization opportunities. All of these choices can affect the size of the intermediate representation and of the final object file at the end of the compilation stage.

In the context of this thesis, the problem of inlining is treated with an offline learning approach, in which vectors of features about call sites and general graph information are collected at the beginning of the compilation stage and are not updated at every step, as an online model would do. The dataset contains C/C++ files from DCMTK, as well as various other public sources, selected by relevance to the inlining problem. Multiple synthetic source files were added in order to extend the dataset. They were generated using a combinatorial approach, ensuring diversity. The full dataset passes through a Clang frontend with the inlining optimization option turned off in order to obtain the corresponding LLVM Intermediate Representation, from which the call sites are extracted and the feature vectors are constructed.

A set of heuristics called teachers is defined and run on feature vectors per program. This is done in order to assign inline/no-inline labels per call site, and the heuristic that manages to reduce the IR most for a program file is chosen for the supervised training of students. The training set is therefore the union of the teachers' labels that performed best per module. The goal is to make the students able to shadow their teachers, a technique called behavior cloning. The choices of the student models are implemented by rewriting call-site sections of the LLVM IR. In the main experiment, a total of 163 C/C++ files were successfully compiled, of which 150 had local call sites and therefore remained in the dataset. The total count of analyzed call sites is 10394. A held-out split was used, where the best model showed an IR reduction of 0.61% compared to never-inline. The best teacher, used as a selected-teacher reference, led to a reduction of 1.10% compared to never-inline. Even though the prototype does not do better than LLVM `-Oz`, it shows that ML policies can be used in inlining optimization instead of relying only on classical heuristic approaches.

Rezumat

În optimizarea compilatoarelor, problema de inlining este de mult studiată pentru a păstra o balanță între dimensiunea aplicațiilor și performanța acestora. Problema alegerii contextului potrivit pentru inlining rămâne complicată, deoarece funcțiile sunt interconectate prin intermediul grafului de apeluri, iar o decizie poate influența structura programului și viitoarele posibilități de optimizare. Toate aceste decizii pot influența dimensiunea reprezentării intermediare, precum și a binarului final, la finalul etapei de compilare.

În contextul acestei teze, problema de inlining este tratată printr-o metodă de învățare offline, în care vectorii de attribute cu informații despre apelurile de funcții și graful de apeluri sunt extrași la începutul compilării, fără a se reactualiza la fiecare pas, precum ar funcționa învățarea online. Setul de date conține fișiere C/C++ din DCMTK și alte surse publice variate, alese pe criteriul relevanței pentru problema de inlining. Multiple surse sintetice au fost adăugate pentru a extinde setul de date și au fost generate combinatorial, pentru a se asigura diversitatea. Întregul set de date trece printr-un frontend Clang cu optimizarea de inlining dezactivată pentru a se obține reprezentarea intermediară LLVM corespunzătoare, de unde se extrag call site-urile și se construiesc vectorii de attribute corespunzători.

Se definește un set de euristici numite profesori, care se rulează pe vectorii de attribute ai fiecărui program pentru a se obține etichete de inline/no-inline per call site. Profesorul care reușește să reducă numărul de instrucțiuni IR cel mai mult pentru un anumit program este folosit în setul pentru training supervizat. Setul de training este alcătuit astfel din reuniunea etichetelor profesorilor care au dus la cea mai mare reducere de IR per modul. Scopul este ca un student să fie capabil să copieze alegerile profesorilor săi, o tehnică numită behavior cloning. Deciziile studenților sunt aplicate prin rescrieri ale reprezentării intermediare la nivelul apelurilor de funcții. În experimentul principal, 163 fișiere C/C++ au fost compilate cu succes, dintre care 150 aveau apeluri locale de funcții și au rămas în setul de date. Numărul total de apeluri analizate este 10394. Pe setul held-out, cel mai bun model a dus la o reducere de IR de 0.61% față de never-inline. Cel mai bun profesor, folosit ca referință selected-teacher, a dus la o reducere de 1.10%. Deși rezultatele nu sunt mai bune decât LLVM -Oz, implementarea demonstrează că politicile de ML pot fi folosite în locul euristiciilor clasice în problema de inlining.

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Chapter 1

Introduction

The problem of choosing when to optimize during compilation remains difficult in its attempt to keep outputs predictable and program runs deterministic across platforms, while also minimizing code size. This has led to a preference for predetermined rules called heuristics, which have long been perfected to give indications about when optimizations would be useful and would give the best results. The issue with normal heuristics is their lack of adaptability, since choosing when to optimize is dependent on multiple factors that keep changing with other optimizations of the call graph. This is why, in this thesis, inlining is looked at through a lens of situational complexity, taking into account how one choice impacts which optimizations will be possible next, the size of the code, and also which other inlining choices can follow [1, 2].

Inlining is one representative example of the above stated compiler problem. Through substituting a call site with the instructions corresponding to the called function, various possibilities for simplification can arise: saving the computational burden of function calls, meaning call overhead, enabling devirtualization, propagating constants, removing dead code, and so on. Although in many cases useful, inlining can lead to the unnecessary expansion of code. To avoid this, various thresholds and cost models are used in production, in order to balance the cases when code increases too much in size and the cases when inlining choices are good for revealing optimization opportunities [1, 2, 3].

In the proposed method in this thesis, the starting point is represented by several deterministic heuristics called teacher policies, on which models referred to as students are trained, with the scope of determining whether they can learn more general rules than their context-specific teachers or give good indicators of when inlining would help. The pipeline keeps track of the inline/no-inline choices of all teachers, which are evaluated on every static call-site feature vector per module. The teacher that results in the smallest size of the rewritten IR is chosen per module, and that corresponding teacher's choices in that module become labels for training. Policies called students are trained on this best-teacher label dataset and learn to imitate the choices of the teachers. The idea of using vectors of features comes from MLGO, the difference being that this prototype implements this behavior-mimicking technique, called imitation learning, offline instead of using the MLGO reinforcement-learning approach [4, 3, 5, 6].

The inlining of functions is a relevant case study, since it impacts both call-level properties and general, global program behavior. The call-level change is the choice to either substitute a call with the callee's contents or to not edit the call site. The global effects appear through the changed structure of the call graph, the size of the caller, the possible redundancy of the callee, as well as new optimization options. This is reproduced in the implementation at a smaller scale: a set of teachers is run on every module, which results in inline/no-inline choices. In the case of the teacher per

module that gives the biggest reduction of the LLVM IR size metric, those choices become labels for the training of binary classifier students.

The `opt` tool in LLVM is used in order to change the LLVM IR, and the number of LLVM IR instructions after rewriting becomes the main metric for student-policy performance, while the size of the native object-file `.text` section is used as a secondary measurement.

1.1 Research Questions and Contributions

The research questions treat inlining not only as a fixed compiler heuristic, but as a decision problem where inline/no-inline choices can be learned from examples. The first goal is to check whether inlining can be formulated as an offline imitation-learning task, where teacher heuristics give labels and student models reproduce their behavior. The second goal is to see whether selecting the best teacher per module, in a BC-Max-style manner, can produce useful labels under a code-size objective.

The contribution is not only the final numerical reduction, but also the full pipeline: static LLVM IR feature extraction, teacher definition and selection, student training, predicted LLVM IR rewrites, and artifact measurement. Table 1.1 summarizes the research questions and corresponding contributions.

ID	Research question	Thesis contribution
RQ1	Can inlining be framed as offline imitation learning from heuristics with a teacher role?	Formalizes inlining as a binary call-site policy problem with teacher heuristics, selected trajectories, and behavior cloning.
RQ2	Can BC-Max-style selection produce call-site labels under a code-size objective?	Evaluates 16 teachers per module and builds selected-teacher labels for supervised training.
RQ3	Can simple students imitate selected teachers from static features?	Implements the LLVM IR workflow for feature extraction, teacher design, behavior cloning, rewriting, size measurement, and student training.
RQ4	Do learned decisions reduce rewritten IR compared with never-inline?	Applies predictions through cloned callees, <code>alwaysinline</code> , LLVM <code>opt</code> , and held-out IR/native <code>.text</code> evaluation.
RQ5	What gaps remain relative to production ML-guided inlining?	Analyzes dataset scale, static rewrite approximation, teacher quality, and missing full LLVM inliner integration.

Table 1.1: Research questions and corresponding contributions.

Chapter 2

Compiler Background

2.1 Compilation Pipeline

The goal of compilation is to obtain a representation of code that can be executed on a machine. The intermediate representation, LLVM IR, and its corresponding instruction count are analyzed in this thesis because they are affected by inlining choices while still being more general than machine code. LLVM IR is generated before backend-dependent decisions, so it is useful for studying optimization choices before the final target-specific code is emitted. A simple version of an LLVM compilation pipeline is:

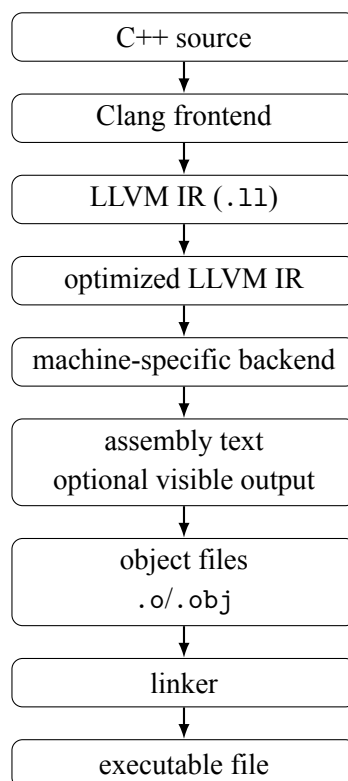


Figure 2.1: Simplified LLVM-based compilation pipeline.

Clang is used for checking syntax and converting C/C++ files into LLVM IR form. Following the optimization pass, the backend makes machine-dependent decisions, such as allocating virtual registers in the IR to real registers and writing the machine instructions that emit object files [7, 8]. LLVM IR is helpful for this implementation because it is more abstract than machine code but still

lower-level than source code.

2.2 LLVM and LLVM IR

LLVM is a project for compiler infrastructure that includes optimization and backend tools, as well as libraries. LLVM IR is just one part of the project, corresponding to LLVM's implementation of an intermediate representation [9, 10]. It has an abstract version of register allocation called virtual registers, which get allocated at the backend stage, and also operations per variable type, memory-level operations, branches, and calls.

2.3 Static and Dynamic Measurements

In the context of this thesis, two concepts are used: static and dynamic call graphs, as well as static and dynamic code size. The static call graph contains all possible call choices, regardless of inputs, while the dynamic call graph is input-dependent and results from one execution of the code. Static code size refers to the dimension of a generated intermediate or final compilation representation, while dynamic code size is a measure of the executed instructions [7, 1].

Chapter 3

Function Inlining

Although inlining looks simplistic at a superficial level, when analyzing the problem in depth, it can be seen that one inline choice, through integrating the callee into the caller's context, can help further compact and optimize code by using known values or context-specific simplifications. The original call instruction may disappear, call overhead can be avoided, and later passes can optimize values and control flow across the merged code region [7, 8, 1].

3.1 Cases Where Inlining Helps

The examples given below were written for the purpose of this thesis. Small auxiliary functions can be simplified easily when inlined:

```
int ab(int a, int b) {                int sum_ab(int a, int b) {
    return a * b;                      return a + b + ab(a, b);
}
```

Figure 3.1: Small helper and caller before inlining.

Through inlining, the call to `ab` is substituted with the multiplication returned by the function. This can improve the computation time needed for changing the call context, called call overhead, if `sum_ab` is executed frequently, meaning it is in a hot path.

When the function call contains constants, inlining can eliminate unused cases, sometimes by propagating constants. In the example below, inlining shows that `c` is 0 and minimizes code size since the `c == 1` branch can be eliminated:

```
// before:
string light_color(int c, int i) {
    if (c == 0)
        return light_red(i);
    if (c == 1)
        return light_green(i);
    return light_blue(i);
}

string car_color(int i) {
    int f = i;
    return light_color(0, 2 + f);
}

// after:
string car_color(int i) {
    int f = i;
    return light_red(2 + f);
}
```

Figure 3.2: Constant-propagation example before and after inlining.

3.2 Cases Where Inlining Hurts

Most recursive cases lead to big increases in code size, which is why most optimization algorithms penalize inlining here:

```
int sum(int x) {                                     // recursive inlining:
    if (x <= 0)                                       // exposes another recursive call
        return 0;                                    // repeated expansion
    return x + sum(x - 1);                           // needs recursion/growth checks
}
```

Figure 3.3: Recursive call site and the reason it is usually rejected.

When a sequence in a program rarely gets executed, inlining can cause unnecessary growth of code:

```
void game_engine_reinit() {                          void entry(bool error) {
    // large recovery path                            if (error)
}                                                    game_engine_reinit();
                                                    }
```

Figure 3.4: Inlining a large block of instructions in a rarely executed path puts pressure on the instruction cache and grows the static code unnecessarily.

When a bigger function gets called many times, its inlining can also expand static code significantly:

```
int get_info(int id) {                               void info_candidates() {
    int s = id;                                       cout << get_info(12) << endl;
    for (int i = 0; i < 1000; ++i)                  cout << get_info(133) << endl;
        s = (s * 33) ^ i;                           cout << get_info(144) << endl;
    return s;                                         // can grow code unnecessarily
}                                                    }
```

Figure 3.5: Large callee copied at multiple call sites if inlined aggressively.

Chapter 4

Problem Formulation

Consider P a program or module, and let

$$C(P) = \{c_1, c_2, \dots, c_n\}$$

be the collection of call sites collected in the order of parsing the program at inlining time. A call site represents places where functions are called in a program, and for each call site c_i , a feature vector x_i is extracted during compilation, with data about the caller, the function which initiates the call, and the callee, the function being called, as well as information about the call graph:

$$x_i = \phi(c_i),$$

where ϕ is the function that gathers features. In the context of the thesis, the features are static and composed of information about the caller and callee, such as their structure, length, functions that call them, how many constants they have, whether they are recursive, as well as some indices regarding the call graph.

In inlining, A is the binary set of possible actions:

$$A = \{0, 1\}, \quad 0 = \text{do not inline}, \quad 1 = \text{inline}.$$

To give insight into how good an action might be, a policy is used that gives a probability of action a for the vector of features x :

$$\pi_\theta(a \mid x)$$

where $\pi_\theta(a \mid x)$ denotes the likelihood of choosing action a , given feature vector x . Then, using $\arg \max$, the inlining choice is made:

$$d_\theta(x) = \arg \max_{a \in A} \pi_\theta(a \mid x).$$

Inlining gives the problem a chronological and interdependent nature. By applying policy π to program P , the following trajectory is output:

$$\tau_\pi(P) = ((x_1, a_1), (x_2, a_2), \dots, (x_T, a_T)),$$

where T represents how many inline/no-inline choices should be made for P . As previously discussed, through the possibility of affecting later optimizations, the decisions are dependent on each other [3, 6].

In the presented prototype, the vector of features is static and collected at the initial compilation stage from LLVM IR. This is done in an offline fashion, because the information is not updated after every inlining choice or change of the call graph, as an online inliner would do. The labels are predicted on this raw data and then the call sites are changed in LLVM IR, without runtime information.

The main scope is minimizing the size of the IR. In order to measure how well policies perform in the implementation, the count of IR instructions after rewriting is used:

$$\text{Size}_{IR}(P, \pi), \quad R(P, \pi) = -\text{Size}_{IR}(P, \pi).$$

The minus sign in the reward function is used because smaller values are desired.

Let

$$\mathcal{H} = \{h_0, h_1, \dots, h_k\}, \quad h^*(P_j) = \arg \min_{h_i \in \mathcal{H}} \text{Size}_{IR}(P_j, h_i)$$

be the set that includes all policies that act as teachers, and $h^*(P_j)$ is the selected teacher for a module P_j , which makes its IR size smallest. Every teacher heuristic outputs an inline/no-inline choice starting from information per call site.

The path taken by the best teacher gets included in the training set for the students:

$$\mathcal{D} = \bigcup_j \{(x_{j,t}, a_{j,t}^*)\}_{t=1}^{T_j},$$

where $a_{j,t}^*$ is the action that the chosen teacher policy $h^*(P_j)$ decided on for the vector of features $x_{j,t}$.

The new policy called a student should theoretically aim to predict labels as closely as possible to its teachers, therefore minimizing a disagreement function:

$$\min_{\theta} \sum_{(x,a) \in \mathcal{D}} \mathbf{1}[\pi_{\theta}(x) \neq a], \quad \mathcal{L}(\theta) = - \sum_{(x,a) \in \mathcal{D}} \log \pi_{\theta}(a \mid x).$$

The practical implementation uses a more direct approach: standard classifiers. For probabilistic classifiers, this can also be written through binary cross-entropy in order to obtain a probability.

The newly trained student can be used as a new policy and added to the teacher set. It gives inline/no-inline predictions per feature vector corresponding to a call site:

$$\hat{a} = \arg \max_{a \in \{0,1\}} \pi_{\theta}(a \mid x).$$

What the trained model predicts leads to rewrites in LLVM IR, and the number of instructions after rewriting is used as the performance metric.

Chapter 5

Related Work

5.1 Classical and Empirical Views of Inlining

Before ML models, this difficult problem was tackled using classical inliners. These attempted to find a compromise between code expansion and the chance for future code simplifications when choosing whether or not to remove call sites. Theodoridis, Grosser, and Su discuss how complex optimal inlining becomes when weighing the impact on consecutive chains of function calls, code size, and following optimization passes [2]. Godbolt gives a similar interpretation for the use of inlining: it can merge code by using known values, eliminate cases that are not needed, and simplify the callee with information known in the caller [1, 11].

5.2 Autotuning and Pre-MLGO Compiler Learning

The idea of empirical search, as well as learning approaches, preceded MLGO. MILEPOST GCC, for instance, constructs vectors of features in order to make the optimization process more adaptive [4]. This was helpful in the methodological sense for the thesis, since the concept of feature vectors was reproduced, and it demonstrated that empirical data can be used for the training of policies, even though its focus was general compiler optimization, not inlining in particular.

The issue of having fixed predetermined parameters in inlining rules was tackled by Cavazos and O’Boyle. They showed useful results by using empirically adjustable parameters in place of static ones [12]. The problem they approached is similar to the one proposed here, but with another scope, since they tried to tune policies automatically, while here the policy is trained on other policies. The technical difference is therefore that autotuning tries to find more appropriate parameters for heuristics, while behavior cloning attempts to train a decision rule starting from feature vectors at call-site level.

The focus on code size, specifically LLVM IR, is intentional in this thesis. It is an easy-to-calculate and objective metric. Object size, also easy to calculate, is listed secondarily. In order to assess real-world performance, other metrics such as runtime have been attempted in the literature, despite their difficulty to measure objectively and their dependence on inputs and target platforms. In recent work, MLGOPerf uses ML techniques modified toward minimizing both code size and performance metrics [13].

5.3 MLGO

MLGO was the initial inspiration for this thesis, because of the idea of using trained models and their predictions for modifying LLVM IR with the purpose of minimizing size [3, 5]. The reinforcement-learning model in MLGO uses states composed of call-graph indicators and call-site data, actions which can be either inlining or not, and a reward showing how well the final compiled program improved, usually through native code-size reduction. Using feature vectors avoids the computational burden of capturing all characteristics of the call graph. MLGO applies a policy-gradient technique, which is expensive because each change needs to be recompiled and the reward can be calculated only at the end [3].

5.4 Offline Imitation Learning and BC-Max

Marinov, Agarwal, and Trofin show that computationally expensive online reinforcement learning is not the only well-performing approach for this problem. They propose using an offline model trained on predefined teachers through imitation learning [6]. For the scope of inlining, let:

$$\mathcal{H} = \{h_0, h_1, \dots, h_n\}$$

represent the collection of heuristic teachers, and let P_i be a module with corresponding $\text{Size}_{IR}(P_i, h)$, meaning the size of the rewritten IR with teacher h .

The selected teacher per module has the smallest size of this rewritten IR:

$$h_i^* = \arg \min_{h \in \mathcal{H}} \text{Size}_{IR}(P_i, h).$$

The trajectory corresponding to the selected heuristic’s choices is the following:

$$\tau_i = \{(s_1, h_i^*(s_1)), \dots, (s_m, h_i^*(s_m))\}.$$

Here, s_t corresponds to the feature vector for call site number t . The learning is done in a supervised way, from the labels of the union of the selected teachers, rather than by means of exploration and recompiling.

What teachers label individual call sites is context-dependent and sometimes does not generalize. Two different teachers might assign different labels to similar call sites in different modules, leading to disagreements in the set used for training the behavior-cloning students. This is why the actual rewrite impact is the real measure of success, not only how well the students agree with teachers.

5.5 Positioning Against Prior Work

Table 5.1 summarizes how the prototype borrows the policy-learning view from MLGO and BC-Max while remaining smaller and more inspectable than production compiler infrastructure.

Work	Main idea	Evaluation focus	Relation to this thesis
Cavazos and O’Boyle [12]	empirical tuning of inliner parameters	benchmark performance	early evidence that inlining choices can be optimized empirically
MILEPOST GCC [4]	program-feature-based optimization selection	empirical benchmark performance	broad compiler ML rather than call-site-level inlining
MLGO [3, 5]	learned policies integrated inside LLVM	code size and performance on large corpora	main learned-heuristic reference, not reproduced here
BC-Max [6]	offline imitation learning from multiple baselines	final objective value	closest template for teacher selection and behavior cloning
Theodoridis et al. [2]	autotuned search for high-reduction inlining-for-size decisions	native binary size on SPEC2017, LLVM, and SQLite	motivates code-size evaluation and stronger baselines
This thesis	BC-Max-style supervised pipeline over static call-site features	rewritten IR and native text-section reference	inspectable offline pipeline for call-site-level policy experiments

Table 5.1: Related-work positioning for ML-guided and empirical inlining optimization.

Chapter 6

Methodology

The implementation follows Figure 2.1 until the output LLVM IR step, whose size is the main objective to minimize. For every individual call site, a feature vector is produced, the set of teachers is run to select the best one per module, students are trained on those labels, and the labels predicted by the student classifiers are reapplied to the IR using LLVM opt. How sources are collected, how IR is generated, how teachers and students are implemented, how IR is rewritten, and how IR size is determined remain visible in the repository.

The dataset contains C/C++ files from various sources, including a small DCMTK snapshot¹, 76 supplements from public sources selected with inlining-relevance criteria, and synthetic sources generated deterministically from a $4^6 = 4096$ configuration space. For the generated sources, the format, content, and size of callees and callers can vary, as well as the order of calls and the arguments. This creates different function sizes, branches, mathematical functions, recursion, constants, and variables. The 70 generated modules contain 1120 call sites, so teachers are run on both representative inline and no-inline scenarios. Extending transferability beyond this generated set remains future work.

Files from public sources are filtered using the criteria of successful compilation to LLVM IR, the existence of a minimum of one local call site, and the ability to pass the teacher-selection step and the corresponding IR rewriting process. These steps happen at the data-collection stage, before training the students. The features in the prototype represent local-level approximations of the publicly available features used by MLGO: information about the call site, the caller and callee, and the general program-level call graph, with flags for recursive functions, constants, number of calls, and number of instructions.

¹DCMTK source repository: <https://github.com/DCMTK/dcmktk>.

Feature group	Implemented fields	Type	What the values indicate
Caller shape	caller-prefixed counts: basic blocks, conditional blocks, users, IR instructions, and call instructions	numeric	size and branching of the function that receives the inlined body; large or call-heavy callers may already be expensive to grow
Callee shape	callee-prefixed counts: basic blocks, conditional blocks, users, IR instructions, and call instructions	numeric	estimated body copied by inlining, plus whether the callee has few users and may become removable after all useful calls are inlined
Call-site context	callsite_height, cost_estimate, and number of constant parameters	numeric	local decision context: approximate graph depth, proxy inline cost, and constants that may enable simplification after the callee body is copied into the caller
Module graph	module edge count and node count	numeric	rough module-level density, used to distinguish sparse files from call-heavy modules where one decision can interact with many nearby calls
Safety signal	recursive-call indicator	binary	marks recursive expansion patterns, where repeated inlining can grow code unnecessarily

Table 6.1: Grouped feature schema used by all teachers and students. The complete ordered feature list is fixed in `scripts/data.py`.

6.1 Dataset Summary

The dataset is composed in order to test the inlining pipeline on more than one type of source: a DCMTK-based source snapshot, generated modules with controlled inline/no-inline cases, and 76 public-source files selected because they compile successfully and expose local call-site information for the feature extractor.

Quantity	Value
Source mix	DCMTK, generated, 76 public files
Candidate C/C++ files	164
Compiled C/C++ files	163
Modules with local call sites	150
Extracted call sites	10394
Train samples	9009
Hold-out samples / modules	1385 / 26
Teacher policies / students	16 / 7
Primary metric	rewritten IR instructions
Secondary metric	native .text size

Source group	Teacher IR red.	Student IR red.
Generated	8.76%	2.75%
Real source	1.22%	1.08%
All modules	1.66%	1.18%

Table 6.2: Dataset summary and source-group IR reduction. “Teacher” means the selected-teacher reference; “Student” means the best full-set student conservative_forest. The most frequent selected teachers are `small_callee` on generated modules and `greedy_ir_size` on real-source modules.

The resulting run starts from 164 candidate C/C++ files. Of these, 163 compile successfully, and 150 remain as LLVM modules with local call sites, giving 10394 call-site samples. The split is done by module, not by individual call site, so nearly identical call contexts from the same file do not appear both in training and testing. At source-group level, generated modules show the largest reduction because they were designed to contain clearer inlining patterns, while real-source files often contain fewer profitable local call sites under the simplified rewrite objective.

Candidate group	Files	Compiled	Reason
DCMTK snapshot	18	17	source-level local call sites
Generated modules	70	70	controlled inlining and no-inlining patterns
Public sources	76	76	compiled C/C++ files with local call-site signal

Table 6.3: Dataset accounting. Filtering is applied before student training and keeps source-level modules with local call sites relevant to the rewritten-IR objective. Public-source provenance, licenses, and selected-file metadata are kept in the repository manifests and source snapshot files.

6.2 Teacher Heuristics

In this thesis, the teacher heuristics are deterministic policies that decide whether a call site should be inlined or kept. They are not production inliners, but inspectable rules for producing behavior-cloning labels. All teachers use the same LLVM IR feature vector, but prioritize different signals: callee size, users, constants, estimated growth, recursion, and call-graph context. Let s be callee size, u callee users, b branch count, k constant arguments, and r a recursion flag. Most rules use the following general shape:

inline if $B(x) - C(x) > 0$, $B(x) \approx 2k + \mathbf{1}_{u=1} + \text{depth signal}$, $C(x) \approx s + b + \text{nested calls} + r$.

The exact thresholds are fixed in `teach.py`; the table gives the readable form of each rule. Most teachers are cheap proxy policies: some prefer small callees, some prefer single-use callees, some reward constant arguments, and others reject recursive or branch-heavy cases. The measured

teacher, `greedy_ir_size`, is more expensive because it tries candidate rewrites and keeps only those that reduce rewritten IR instruction count.

Teacher	Short rule
<code>never_inline</code>	Conservative baseline: never inline.
<code>small_callee</code>	Inline if s is small and $r = 0$.
<code>single_caller</code>	Inline if $u = 1$ and the callee is not large; the standalone helper may disappear after cleanup.
<code>constant_argument</code>	Inline if $k > 0$, so caller-known constants may remove branches or simplify expressions.
<code>cost_budget,</code> <code>growth_budget</code>	Inline if estimated growth $G \approx s \max(1, u) + b$ + nested calls stays under a budget.
<code>loop_averse,</code> <code>hot_leaf</code>	Prefer leaf-like callees; reject large, branchy, or call-heavy callees.
<code>balanced_score,</code> <code>llvm_like_size,</code> <code>benefit_cost_ratio</code>	Inline if $B(x) / \max(1, C(x))$ passes a threshold.
<code>aggressive_speed</code>	Accept more small and medium calls, while still rejecting recursive or clearly large cases.
<code>ml_linear_score</code>	Inline if $w \cdot x + \beta > 0$, using fixed hand-written weights rather than training.
<code>bandit_ucb_proxy</code>	Inline if $\hat{g}(x) + \alpha / \sqrt{1 + u} > 0$: estimated gain plus a small uncertainty bonus.
<code>rl_value_proxy</code>	Inline if immediate simplification plus future call-graph value exceeds growth penalty.
<code>greedy_ir_size</code>	Measured teacher: order plausible non-recursive candidates, try candidate rewrites, and keep a candidate only when the actual rewritten IR instruction count decreases.

Table 6.4: Teacher policies as readable decision rules.

This distinction matters because the student does not learn from a universal oracle that is always correct for every call site. For every module, the system selects the teacher that gives the smallest rewritten IR, and that teacher’s labels enter the training dataset. The dataset is therefore the union of decisions made by different teachers, each chosen because it performed best for one module. The main run uses one offline teacher-selection round; Chapter 8 also reports a three-round check where the best learned student is inserted back into the measured policy set.

Chapter 7

Implementation

The implementation is organized as a reproducible pipeline, starting from source files and ending with rewritten LLVM IR artifacts. The scripts generate synthetic modules, compile C/C++ sources to LLVM IR, extract call-site features, evaluate teacher policies, train student models, rewrite selected call sites, and produce the measurements and plots used in the thesis. The repository keeps the source collection, generated files, intermediate IR, model decisions, and final results inspectable:

<https://github.com/dirgnic/optimizingInlining>

7.1 Repository Structure and Execution

The pipeline can be run in two main ways. The full run, `python3 scripts/run.py all`, executes the experiment and produces the reported numbers. The smaller demo run, `python3 scripts/run.py demo`, compiles a few generated modules, prints the call-site order and inline/-keep decisions, and writes before/after LLVM IR files under `out/demo/rewritten/`, making the rewrite step inspectable without the full dataset.

The scripts follow the fixed execution order in Table 7.1: define shared data structures, generate and compile sources, parse LLVM IR, extract local call-site features, apply teacher rules, select the best teacher per module, build the behavior-cloning dataset, train students, apply predictions through the rewrite path, measure rewritten IR and native text-section sizes, and generate summaries and figures.

Step	Script	Role
0	<code>data.py</code>	defines shared directories, feature names, and call-site data structures.
1	<code>run.py</code>	coordinates the full command <code>python3 scripts/run.py all</code> and the short demo command <code>python3 scripts/run.py demo</code> .
2	<code>build.py</code>	generates the deterministic synthetic modules, compiles C/C++ sources to LLVM IR, and measures native compiler baselines.
3	<code>ir.py</code>	parses LLVM IR and extracts feature vectors for local call sites.
4	<code>teach.py</code> and <code>rewrite.py</code>	apply teacher rules, rewrite IR with LLVM opt, and select the best teacher per module by rewritten IR instruction count.
5	<code>teach.py</code>	converts the selected teacher actions into the behavior-cloning dataset.
6	<code>learn.py</code>	trains the student classifiers and stores their inline/no-inline predictions.
7	<code>rewrite.py</code>	applies student predictions through the same clone-and-opt rewrite path, then measures rewritten IR and native text-section size.
8	<code>graph.py</code> , <code>paper.py</code> , and <code>visual.py</code>	write the call-graph trace, generated Markdown summary, and figures used for inspection and reporting.

Table 7.1: Default script order for the reported thesis pipeline.

The local environment for the reported run is macOS on Apple Silicon, Homebrew Clang/LLVM 15.0.7, and Python 3.9.6. Sources are compiled with inlining disabled, so the call graph can be observed before normal inlining optimizations are applied. Deterministic generation, fixed model seeds, and a deterministic module-level split support reproducibility.

7.2 Rewrite Algorithm

The rewrite algorithm works at call-site level. When a call site is selected for inlining, the callee is cloned, the selected call is redirected to that clone, and LLVM opt performs inlining and cleanup. Recursive, unresolved, or unsafe calls are skipped. This is not LLVM’s production inliner, but a smaller mechanism built so predicted inline/no-inline decisions can be materialized and measured.

The order of call sites is deterministic. Modules are processed in sorted LLVM IR filename order, functions are visited in the order in which their `define` blocks appear, and calls are scanned from top to bottom inside each function. This is not dynamic execution order and not a topological ordering of the call graph. It is only the textual order used by the extractor. The measured `greedy_ir_size` teacher is the exception, because it reorders plausible candidates by an estimated benefit/cost score before trying rewrites.

Student model	Role in the experiment
<code>logistic_regression</code>	Linear, interpretable classifier.
<code>decision_tree</code>	Small threshold tree over individual features.
<code>knn</code>	Copies labels from nearby feature vectors.
<code>random_forest</code>	Ensemble of shallow trees.
<code>deep_forest</code>	Forest with deeper trees, giving a more flexible rule set.
<code>conservative_forest</code>	Forest variant with a higher effective inline threshold.
<code>small_mlp</code>	One-hidden-layer neural classifier.

Table 7.2: Student model families used for behavior cloning.

The student models are standard supervised classifiers. They do not explore random inlining choices or recompile programs during training. They receive the same static feature vectors and try to reduce disagreement with the selected-teacher labels. The models are intentionally simple: logistic regression gives a linear baseline, trees and forests represent threshold-based rules, KNN checks nearby call sites, and a small MLP adds a lightweight neural model. Their purpose is to test whether selected teacher behavior can be imitated from static call-site information.

Chapter 8

Evaluation

The evaluation measures two aspects of the system: teacher-label match, which shows whether a student imitates selected-teacher labels on held-out call sites, and the actual size effect after applying predicted rewrites. This distinction matters because high label match does not automatically mean smaller rewritten IR; a few mistakes can be expensive if a student incorrectly inlines a large or risky callee.

In the reported run, 163 C/C++ sources compile successfully and produce 150 LLVM modules with local call sites, from which 10394 call-site samples are extracted. The split is performed by module: 9009 samples from 124 modules are used for training, and 1385 samples from 26 modules are kept for held-out testing. This avoids testing on call sites from the same module that was already used during training.

During teacher evaluation, all teacher policies are run on every module. For each module, the selected teacher is the one that gives the smallest rewritten IR size. On held-out modules, this selected teacher is an oracle-style reference because it is chosen after all candidate teachers are measured; it is not a single policy deployable without the selection step. Across all 150 modules, the selected-teacher reference reduces average rewritten IR from 1985.39 to 1952.39 instructions, a 1.66% reduction. On the held-out split, it reduces IR by 1.10%, from 1569.23 to 1552.04 average instructions.

Model	Teacher-label match	Held-out Avg. IR	IR reduction	Gap to teacher
logistic_regression	0.901	1569.69	-0.03%	17.65
decision_tree	0.944	1565.50	0.24%	13.46
knn	0.914	1588.08	-1.20%	36.04
random_forest	0.952	1562.46	0.43%	10.42
deep_forest	0.955	1559.62	0.61%	7.58
conservative_forest	0.947	1562.69	0.42%	10.65
small_mlp	0.931	1607.35	-2.43%	55.31
deep_forest per-module: median 0.17%, min -0.52%, max 21.76%, 18 improved, 7 same, 1 worse.				

Table 8.1: Held-out student results on the 26 test modules. Teacher-label match measures imitation of selected-teacher labels; IR reduction and gap to selected teacher measure the rewritten-IR result.

The results show that students can learn part of the selected-teacher behavior, but they do not reach the selected-teacher reference. This is expected because the selected labels come from multiple teachers and may contain context-dependent decisions. A model can imitate many labels correctly while still making a few costly mistakes that grow IR, which is why both label match and

rewritten IR size are reported.

The student families, thresholds, and hyperparameters are fixed in the scripts before this evaluation. The phrase “best held-out student” is descriptive: it reports which fixed model had the smallest held-out rewritten IR in this run. No additional hyperparameter search or model-family selection is performed on the held-out split.

Policy	Held-out Avg. IR	IR reduction
never_inline	1569.23	0.00%
attempted always-inline	1995.27	-27.15%
selected teacher	1552.04	1.10%
deep_forest	1559.62	0.61%
conserv. forest	1562.69	0.42%

Table 8.2: Held-out reference comparison.

Native workflow	.text bytes	Obj. bytes
no_inline_0z	674660	2225096
llvm_0z	574044	1585120
Reduction	14.91%	28.76%

Table 8.3: Compact LLVM -0z native-size comparison.

Scope	Teacher IR red.	Student IR red.
All 80	1.22%	1.08%
13 held-out	0.32%	0.19%

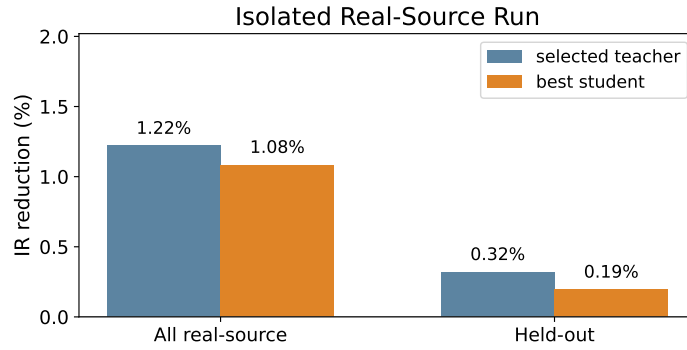


Table 8.4: All-real-source isolated run. The experiment retrains and rewrites only real-source modules; the plot visualizes the same reductions as the table.

For the student models, the best held-out rewritten-IR result is obtained by `deep_forest`: 1559.62 average IR instructions, or a 0.61% reduction relative to `never-inline`. On the full artifact set, `conservative_forest` is best, reaching 1961.94 average instructions and a 1.18% reduction. The reductions are positive but modest, especially on the held-out split.

The LLVM -0z comparison shows the distance from production optimization: it is much stronger on standalone C/C++ files, reducing native .text size and object-file size by a larger margin. The learned policies should therefore be read as a prototype and methodology result, not as a replacement for LLVM’s full size-optimization pipeline. In the additional three-round BC-Max-style check, feeding the best student back into the measured policy set does not improve held-out results: the reference stays at 1.10% and `deep_forest` remains best at 0.61%.

Chapter 9

Limitations

- **Dataset scale.** One of the most important limitations is the scale of the dataset. In the main experiment, 163 C/C++ files were successfully compiled, and 10394 call-site samples were extracted, including examples from 76 public-source files. Even though this is enough for the implemented prototype, stronger generalization would require larger benchmark families, such as SPEC, cBench/MiBench, SQLite builds, and wider LLVM test-suite subsets.
- **Held-out scope.** The held-out split contains 1385 samples from 26 unseen modules. However, all modules still come from the same general source collection and generator regime. A stronger future evaluation would use modules from more clearly different sources, in order to test whether the learned students generalize outside the current dataset composition.
- **Rewrite approximation.** In the main experiment, the feature vectors are extracted once from the initial LLVM IR. The chosen call sites are then rewritten by cloning selected callees, marking them with `alwaysinline`, and letting LLVM opt perform the actual inlining and cleanup. This is an offline approximation: the call graph and the feature vectors are not recomputed after every accepted decision, as they would be in a fuller online inlining system closer to LLVM’s own inliner.
- **Metric scope.** The main scope of the prototype is minimizing rewritten LLVM IR instruction count. Native `.text` size and object-file size are used as secondary measurements, but final linked executable size and runtime behavior are outside the current evaluation. Therefore, the results should be interpreted as IR-level and artifact-level evidence, not as a complete performance evaluation.
- **Teacher and MLGO scope.** BC-Max-style selection can only choose among the available candidate policies. In this thesis, the teachers are deterministic and inspectable proxy heuristics, not production inliner policies. A full reproduction of MLGO would require deeper integration into LLVM, stronger learned or production-level teachers, and a much larger corpus.
- **Iterative scope.** A three-round BC-Max-style check was also attempted, where the best learned student was fed back into the set of measured candidate policies. This did not lead to larger improvements in the reported run. Moreover, this check does not include the uncertainty-based trajectory exploration used in larger BC-Max-style systems.
- **Generated-data filtering.** The generated modules are useful because they add diverse and controlled inlining cases, but they are still synthetic. A future generator could add a discriminator that rejects near-duplicate modules and keeps rarer or more informative inlining patterns before training.

Chapter 10

Conclusions

The purpose of this thesis was to study how compiler inlining can be optimized using offline methods for learning policies, and to implement a reproducible end-to-end pipeline that works with LLVM IR. In the reported run, 163 C/C++ sources are compiled successfully, 150 LLVM modules with local call sites are evaluated, 16 teacher heuristics are run, 7 student models are trained, and the predicted inline/no-inline decisions are applied back to LLVM IR through call-site-level rewrites.

The main held-out result is measured on a source-level mix composed of DCMTK files, generated modules, and 76 public C/C++ source files. On the 26 held-out modules, the selected-teacher reference reduces the average rewritten LLVM IR instruction count from 1569.23 to 1552.04, which corresponds to a 1.10% reduction relative to the never-inline teacher. For the student models, the best held-out result by IR size is obtained by `deep_forest`, which reaches 1559.62 average IR instructions, corresponding to a 0.61% reduction. On the full artifact set, `conservative_forest` reaches 1961.94 average instructions and reduces rewritten IR by 1.18%.

The comparison with LLVM `-Oz` shows that the prototype should not be interpreted as a production-ready replacement for LLVM’s full optimization pipeline. On the standalone C/C++ files, LLVM `-Oz` reduces native `.text` size by 14.91% and object-file size by 28.76% relative to `no_inline_Oz`. The learned policies therefore demonstrate the workflow and problem formulation, rather than outperforming LLVM’s production size pipeline.

The major contribution of the thesis is methodological. The implementation connects source collection, LLVM IR generation, feature extraction, teacher evaluation, behavior-cloning dataset construction, student training, rewrite application, and artifact-level measurement in one reproducible workflow. The results show that call-site-level inlining can be framed as an offline policy-learning problem, where students imitate the best available teacher trajectories and their predictions are then tested through actual LLVM IR rewrites. Stronger future results would require integration with LLVM’s real inliner, larger real benchmark suites, stronger teacher policies, and measurements on final linked binaries.

Appendix A

Additional Diagnostics

The appendix contains diagnostic plots that support the main results. They show which teachers contribute labels, how generated and real-source modules differ, how well students imitate selected labels, and whether student decisions become too aggressive or too conservative after LLVM IR rewriting.

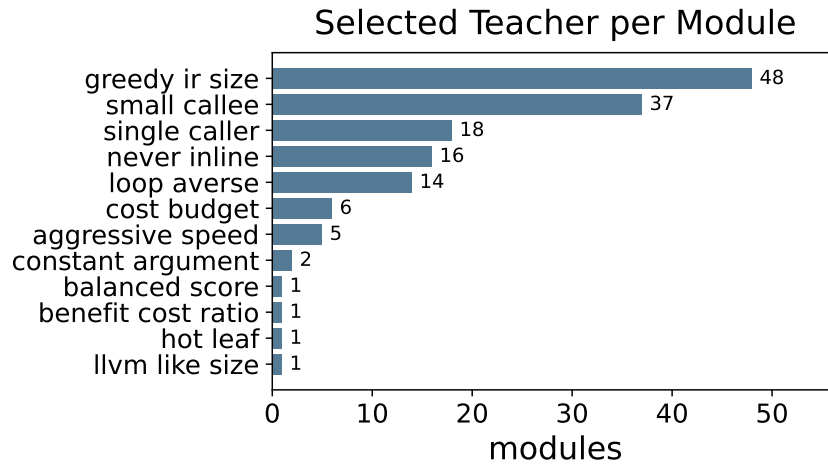


Figure A.1: Selected teacher distribution by module.

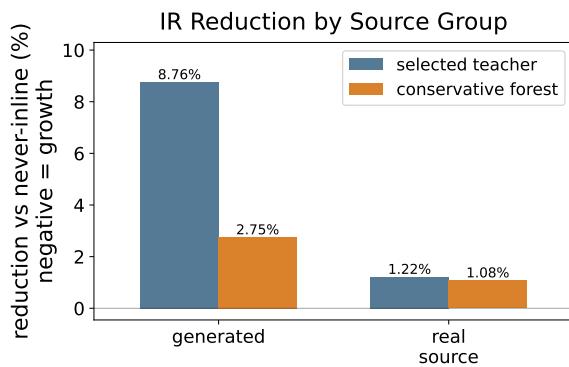


Figure A.2: Teacher and student IR reduction by source group.

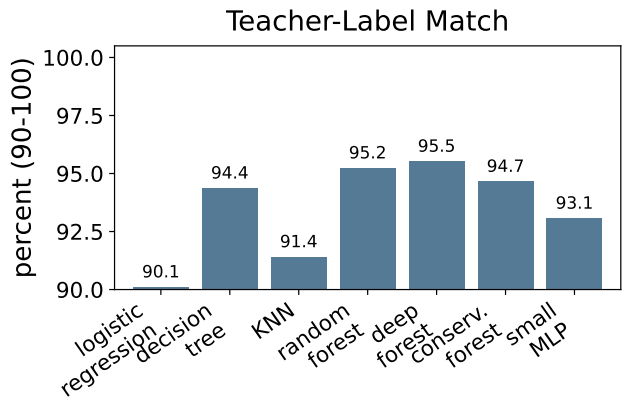


Figure A.3: Held-out teacher-label match for student models.

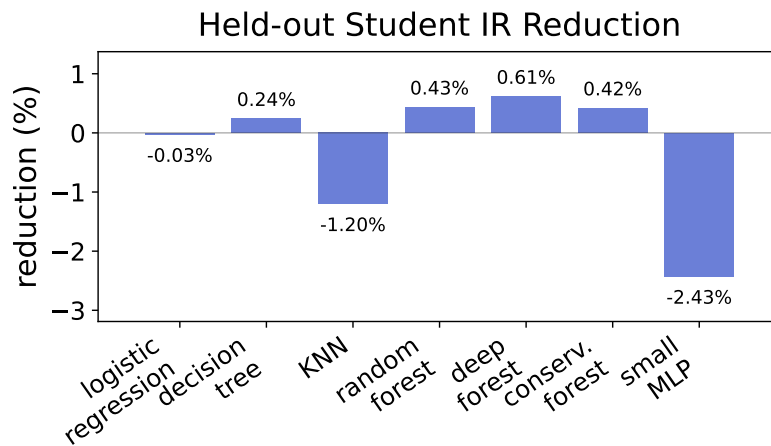


Figure A.4: Held-out student IR reduction.

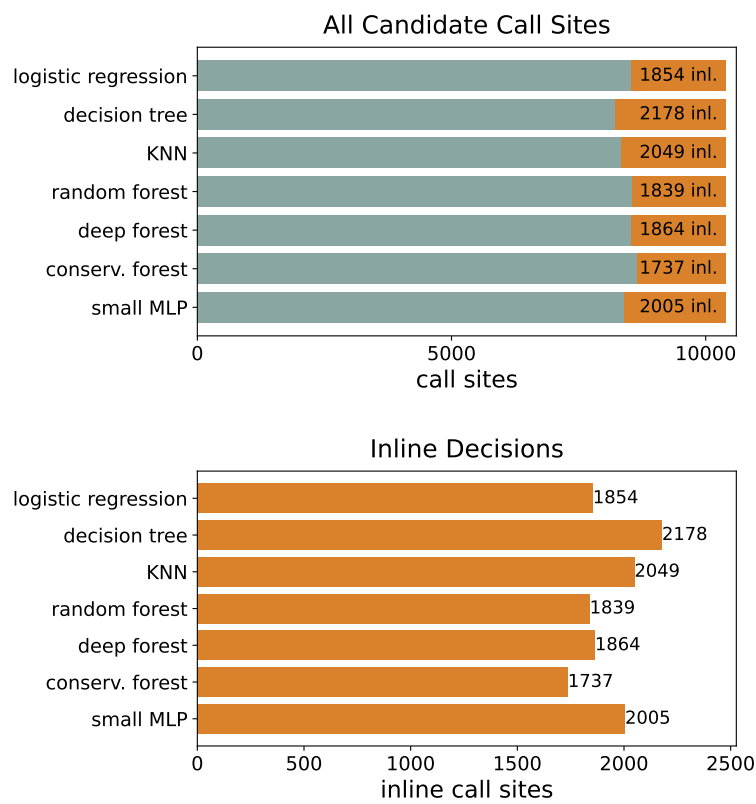


Figure A.5: Student inline/no-inline decisions across candidate call sites.

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